CBSE Class 12 Mathematics (Delhi 2008)

Solutions and Computational Verification Manual

G. V. V. Sharma
Department of Electrical Engineering
Indian Institute of Technology Hyderabad

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ABOUT THIS DOCUMENT

This document provides complete step-by-step analytical solutions to the CBSE Class 12 Mathematics 2008 Delhi Board Examination (29 questions across Sections A, B, and C). In accordance with modern scientific computing paradigms:

1. Each question is accompanied by theoretical derivations cross-referenced to standard formulae in Section 1.1.
2. Differential equations are solved strictly using **Laplace transform** methods.
3. Vector magnitude problems are solved using the rigorous **norm of vector** ($\|\mathbf{v}\| = \sqrt{\mathbf{v}^\top \mathbf{v}}$) formulation.
4. Every problem includes verified **C** or **Python** scripts located in the `codes/` directory, designed for compilation and execution in mobile Linux environments such as Termux.
5. Geometric and analytical curves are plotted with Matplotlib and saved directly as raster figures for inspection.

Contents

1 CBSE Class 12 Mathematics 2008 **3**
1.1 Formulae 3
1.2 Problems and Solutions 5

1 CBSE Class 12 Mathematics 2008

1.1 Formulae

1.1.1 Probability Rules

$$\Pr(A+B) = \Pr(A) + \Pr(B) - \Pr(AB) \tag{1}$$

$$\Pr(A|B) = \frac{\Pr(AB)}{\Pr(B)} \tag{2}$$

$$\Pr(E_i|A) = \frac{\Pr(E_i)\Pr(A|E_i)}{\sum_{k=1}^{n}\Pr(E_k)\Pr(A|E_k)} \quad \text{(Bayes' Theorem)} \tag{3}$$

$$P(X=r) = \binom{n}{r} p^r q^{n-r}, \quad q = 1-p \quad \text{(Binomial Distribution)} \tag{4}$$

1.1.2 Laplace Transforms for Differential Equations

$$\mathcal{L}\{y(t)\} = Y(s) = \int_0^\infty e^{-st} y(t)\,dt \tag{5}$$

$$\mathcal{L}\{y'(t)\} = sY(s) - y(0) \tag{6}$$

$$\mathcal{L}\{y''(t)\} = s^2Y(s) - sy(0) - y'(0) \tag{7}$$

$$\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}, \quad \mathcal{L}\{e^{at}\} = \frac{1}{s-a} \tag{8}$$

$$\mathcal{L}^{-1}\left\{\frac{1}{(s+a)^2}\right\} = te^{-at} \tag{9}$$

1.1.3 Vectors and Norm Procedure

$$\|\mathbf{a}\| = \sqrt{\mathbf{a}^\top\mathbf{a}} = \sqrt{a_1^2 + a_2^2 + a_3^2} \tag{10}$$

$$\hat{\mathbf{a}} = \frac{\mathbf{a}}{\|\mathbf{a}\|} \tag{11}$$

$$\cos\theta = \frac{\mathbf{a}^\top\mathbf{b}}{\|\mathbf{a}\|\|\mathbf{b}\|} \tag{12}$$

$$\mathbf{a}\times\mathbf{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} \tag{13}$$

$$d = \frac{|(\mathbf{a}_2 - \mathbf{a}_1)^\top(\mathbf{b}_1\times\mathbf{b}_2)|}{\|\mathbf{b}_1\times\mathbf{b}_2\|} \quad \text{(Shortest distance between skew lines)} \tag{14}$$

1.1.4 Matrices and Determinants

$$A = \frac{1}{2}(A + A^\top) + \frac{1}{2}(A - A^\top) \tag{15}$$

$$\det(AB) = \det(A)\det(B), \quad \det(A^\top) = \det(A) \tag{16}$$

$$C_{ij} = (-1)^{i+j}M_{ij} \tag{17}$$

1.1.5 Calculus and Trigonometry

$$\tan^{-1} x + \tan^{-1} y = \tan^{-1}\left(\frac{x+y}{1-xy}\right), \quad xy < 1 \tag{18}$$

$$\int \frac{f'(x)}{f(x)}\,dx = \ln|f(x)| + C \tag{19}$$

$$\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx \tag{20}$$

1.2 Problems and Solutions

Section A (1 Mark Each)

1.2.1 Question 1 If $f(x) = x + 7$ and $g(x) = x - 7$, $x \in \mathbb{R}$, find $(f \circ g)(7)$.
Solution: By definition of composite functions,

$$(f \circ g)(x) = f(g(x)) \tag{21}$$

Evaluating at $x = 7$:

$$g(7) = 7 - 7 = 0 \tag{22}$$
$$f(g(7)) = f(0) = 0 + 7 = 7 \tag{23}$$

Hence, $(f \circ g)(7) = 7$.

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codes/q01.c
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1.2.2 Question 2 Evaluate: $\sin\left[\frac{\pi}{3} - \sin^{-1}\left(-\frac{1}{2}\right)\right]$.
Solution: Since $\sin^{-1}(-x) = -\sin^{-1}(x)$, we have:

$$\sin^{-1}\left(-\frac{1}{2}\right) = -\frac{\pi}{6} \tag{24}$$

Substituting this value into the expression:

$$\sin\left[\frac{\pi}{3} - \left(-\frac{\pi}{6}\right)\right] = \sin\left(\frac{\pi}{3} + \frac{\pi}{6}\right) = \sin\left(\frac{\pi}{2}\right) = 1 \tag{25}$$

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codes/q02.c
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1.2.3 Question 3 Find the value of x and y if:

$$2\begin{pmatrix}1 & 3\\ 0 & x\end{pmatrix} + \begin{pmatrix}y & 0\\ 1 & 2\end{pmatrix} = \begin{pmatrix}5 & 6\\ 1 & 8\end{pmatrix} \tag{26}$$

Solution: Performing scalar multiplication and matrix addition:

$$\begin{pmatrix}2 & 6\\ 0 & 2x\end{pmatrix} + \begin{pmatrix}y & 0\\ 1 & 2\end{pmatrix} = \begin{pmatrix}2+y & 6\\ 1 & 2x+2\end{pmatrix} = \begin{pmatrix}5 & 6\\ 1 & 8\end{pmatrix} \tag{27}$$

Equating corresponding entries:

$$2 + y = 5 \implies y = 3 \tag{28}$$
$$2x + 2 = 8 \implies 2x = 6 \implies x = 3 \tag{29}$$

Thus, $x = 3$ and $y = 3$.

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codes/q03.c
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1.2.4 Question 4 Evaluate:

$$\begin{vmatrix} a+ib & c+id \\ -c+id & a-ib \end{vmatrix} \tag{30}$$

Solution: Expanding the 2×2 determinant:

$$\Delta = (a+ib)(a-ib) - (c+id)(-c+id) \tag{31}$$
$$= (a^2 - i^2b^2) - (-(c^2 - i^2d^2)) \tag{32}$$
$$= (a^2+b^2) - (-(c^2+d^2)) = a^2+b^2+c^2+d^2 \tag{33}$$

codes/q04.c

1.2.5 Question 5 Find the co-factor of a_{12} in the determinant:

$$\begin{vmatrix} 2 & -3 & 5 \\ 6 & 0 & 4 \\ 1 & 5 & -7 \end{vmatrix} \tag{34}$$

Solution: From formula (17), the cofactor C_{12} is given by:

$$C_{12} = (-1)^{1+2} M_{12} = -\begin{vmatrix} 6 & 4 \\ 1 & -7 \end{vmatrix} \tag{35}$$
$$= -(6(-7) - 4(1)) = -(-42-4) = -(-46) = 46 \tag{36}$$

codes/q05.c

1.2.6 Question 6 Evaluate:

$$\int \frac{x^2}{1+x^3}\, dx \tag{37}$$

Solution: Let $t = 1 + x^3$. Then $dt = 3x^2 dx$, which implies $x^2 dx = \frac{1}{3} dt$. From (19):

$$\int \frac{x^2}{1+x^3}\, dx = \frac{1}{3}\int \frac{dt}{t} = \frac{1}{3}\ln|t| + C = \frac{1}{3}\ln|1+x^3| + C \tag{38}$$

codes/q06.py

1.2.7 Question 7 Evaluate:

$$\int_0^1 \frac{dx}{1+x^2} \tag{39}$$

Solution: Using the standard integral $\int \frac{dx}{1+x^2} = \tan^{-1} x$:

$$\int_0^1 \frac{dx}{1+x^2} = \left[\tan^{-1} x\right]_0^1 = \tan^{-1}(1) - \tan^{-1}(0) = \frac{\pi}{4} - 0 = \frac{\pi}{4} \tag{40}$$

codes/q07.c

1.2.8 Question 8 Find a unit vector in the direction of $\mathbf{a} = 3\hat{i} + 2\hat{j} + 6\hat{k}$.

Solution: From formula (10), we evaluate the norm:

$$\|\mathbf{a}\| = \sqrt{\mathbf{a}^\top \mathbf{a}} = \sqrt{3^2 + 2^2 + 6^2} = \sqrt{9 + 4 + 36} = \sqrt{49} = 7 \tag{41}$$

From formula (11), the unit vector is:

$$\hat{\mathbf{a}} = \frac{\mathbf{a}}{\|\mathbf{a}\|} = \frac{3\hat{i} + 2\hat{j} + 6\hat{k}}{7} = \frac{3}{7}\hat{i} + \frac{2}{7}\hat{j} + \frac{6}{7}\hat{k} \tag{42}$$

codes/q08.c

1.2.9 Question 9 Find the angle between the vectors $\mathbf{a} = \hat{i} - 2\hat{j} + 3\hat{k}$ and $\mathbf{b} = 3\hat{i} - 2\hat{j} + \hat{k}$.

Solution: From formula (10), compute the norms:

$$\|\mathbf{a}\| = \sqrt{1^2 + (-2)^2 + 3^2} = \sqrt{1 + 4 + 9} = \sqrt{14} \tag{43}$$

$$\|\mathbf{b}\| = \sqrt{3^2 + (-2)^2 + 1^2} = \sqrt{9 + 4 + 1} = \sqrt{14} \tag{44}$$

The inner product is:

$$\mathbf{a}^\top \mathbf{b} = (1)(3) + (-2)(-2) + (3)(1) = 3 + 4 + 3 = 10 \tag{45}$$

From formula (12):

$$\cos\theta = \frac{\mathbf{a}^\top \mathbf{b}}{\|\mathbf{a}\|\|\mathbf{b}\|} = \frac{10}{\sqrt{14}\sqrt{14}} = \frac{10}{14} = \frac{5}{7} \implies \theta = \cos^{-1}\left(\frac{5}{7}\right) \tag{46}$$

codes/q09.c

1.2.10 Question 10 For what value of λ are the vectors $\mathbf{a} = 2\hat{i} + \lambda\hat{j} + \hat{k}$ and $\mathbf{b} = \hat{i} - 2\hat{j} + 3\hat{k}$ perpendicular to each other?

Solution: Two vectors are orthogonal if and only if their inner product is zero:

$$\mathbf{a}^\top \mathbf{b} = 0 \implies (2)(1) + (\lambda)(-2) + (1)(3) = 0 \tag{47}$$

$$\implies 2 - 2\lambda + 3 = 0 \implies 5 - 2\lambda = 0 \implies \lambda = \frac{5}{2} \tag{48}$$

codes/q10.c

Section B (4 Marks Each)

1.2.11 Question 11 Is the binary operation $*$ defined on $\mathbb{Q}$ by $a*b = \frac{a+b}{2}$ commutative? Is it associative?

Solution:

1. **Commutativity:** For all $a, b \in \mathbb{Q}$:

$$b * a = \frac{b+a}{2} = \frac{a+b}{2} = a * b \tag{49}$$

Hence, $*$ is commutative.

2. **Associativity:** Consider three elements $a, b, c \in \mathbb{Q}$:

$$(a * b) * c = \left(\frac{a+b}{2}\right) * c = \frac{\frac{a+b}{2} + c}{2} = \frac{a+b+2c}{4} \tag{50}$$

$$a * (b * c) = a * \left(\frac{b+c}{2}\right) = \frac{a + \frac{b+c}{2}}{2} = \frac{2a+b+c}{4} \tag{51}$$

Since $(a * b) * c \neq a * (b * c)$ in general, $*$ is not associative.

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codes/q11.c
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1.2.12 Question 12 Prove that:

$$\tan^{-1}\frac{1}{3} + \tan^{-1}\frac{1}{5} + \tan^{-1}\frac{1}{7} + \tan^{-1}\frac{1}{8} = \frac{\pi}{4} \tag{52}$$

Solution: Applying formula (18) pair-wise:

$$\tan^{-1}\frac{1}{3} + \tan^{-1}\frac{1}{5} = \tan^{-1}\left(\frac{\frac{1}{3}+\frac{1}{5}}{1-\frac{1}{15}}\right) = \tan^{-1}\left(\frac{8/15}{14/15}\right) = \tan^{-1}\left(\frac{4}{7}\right) \tag{53}$$

$$\tan^{-1}\frac{1}{7} + \tan^{-1}\frac{1}{8} = \tan^{-1}\left(\frac{\frac{1}{7}+\frac{1}{8}}{1-\frac{1}{56}}\right) = \tan^{-1}\left(\frac{15/56}{55/56}\right) = \tan^{-1}\left(\frac{3}{11}\right) \tag{54}$$

Combining both results:

$$\tan^{-1}\left(\frac{4}{7}\right) + \tan^{-1}\left(\frac{3}{11}\right) = \tan^{-1}\left(\frac{\frac{4}{7}+\frac{3}{11}}{1-\frac{12}{77}}\right) = \tan^{-1}\left(\frac{\frac{44+21}{77}}{\frac{77-12}{77}}\right) = \tan^{-1}\left(\frac{65}{65}\right) = \tan^{-1}(1) = \frac{\pi}{4} \tag{55}$$

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codes/q12.c
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1.2.13 Question 13 Let $A = \begin{pmatrix} 3 & 4 & 5 \\ 4 & 4 & 3 \\ 0 & 4 & 7 \end{pmatrix}$. Express A as the sum of a symmetric and a skew-symmetric matrix.

Solution: From formula (15), $A = P + Q$ where $P = \frac{1}{2}(A + A^\top)$ is symmetric and $Q = \frac{1}{2}(A - A^\top)$ is skew-symmetric.

$$A^\top = \begin{pmatrix} 3 & 4 & 0 \\ 4 & 4 & 4 \\ 5 & 3 & 7 \end{pmatrix} \tag{56}$$

$$P = \frac{1}{2}\begin{pmatrix} 6 & 8 & 5 \\ 8 & 8 & 7 \\ 5 & 7 & 14 \end{pmatrix} = \begin{pmatrix} 3 & 4 & 2.5 \\ 4 & 4 & 3.5 \\ 2.5 & 3.5 & 7 \end{pmatrix} \tag{57}$$

$$Q = \frac{1}{2}\begin{pmatrix} 0 & 0 & 5 \\ 0 & 0 & -1 \\ -5 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 2.5 \\ 0 & 0 & -0.5 \\ -2.5 & 0.5 & 0 \end{pmatrix} \tag{58}$$

Clearly $P^\top = P$, $Q^\top = -Q$, and $P + Q = A$.

codes/q13.c

1.2.14 Question 14 For what value of k is the following function continuous at $x = 2$?

$$f(x) = \begin{cases} 2x + 1, & x < 2 \\ k, & x = 2 \\ 3x - 1, & x > 2 \end{cases} \tag{59}$$

Solution: For continuity at $x = 2$, $\lim_{x\to 2^-} f(x) = \lim_{x\to 2^+} f(x) = f(2)$.

$$\lim_{x\to 2^-} f(x) = \lim_{x\to 2^-} (2x + 1) = 2(2) + 1 = 5 \tag{60}$$

$$\lim_{x\to 2^+} f(x) = \lim_{x\to 2^+} (3x - 1) = 3(2) - 1 = 5 \tag{61}$$

$$f(2) = k \tag{62}$$

Therefore, $k = 5$.

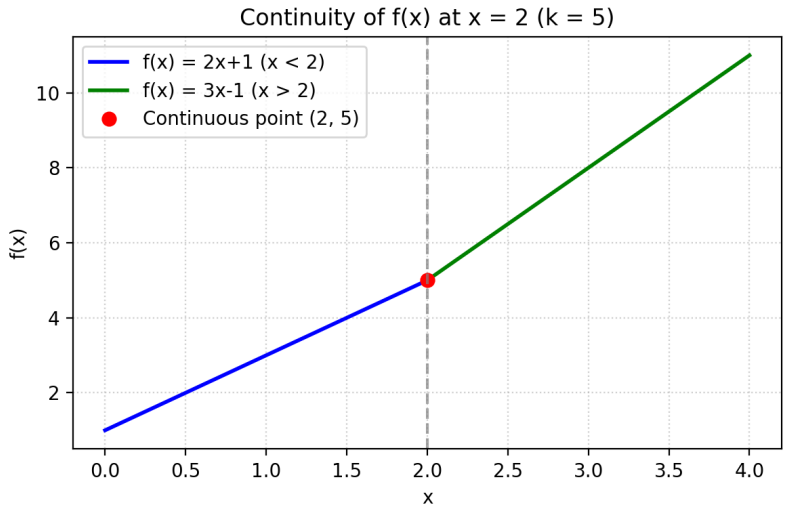


Figure 1: Continuity of $f(x)$ at $x = 2$ for $k = 5$.

codes/q14.py

1.2.15 Question 15 Differentiate with respect to x:

$$y = \tan^{-1}\left(\frac{\sqrt{1+x} - \sqrt{1-x}}{\sqrt{1+x} + \sqrt{1-x}}\right) \tag{63}$$

Solution: Substitute $x = \cos 2\theta$, so $\theta = \frac{1}{2}\cos^{-1} x$.

$$\sqrt{1+x} = \sqrt{1+\cos 2\theta} = \sqrt{2\cos^2\theta} = \sqrt{2}\cos\theta \tag{64}$$

$$\sqrt{1-x} = \sqrt{1-\cos 2\theta} = \sqrt{2\sin^2\theta} = \sqrt{2}\sin\theta \tag{65}$$

Substituting into y:

$$y = \tan^{-1}\left(\frac{\sqrt{2}\cos\theta - \sqrt{2}\sin\theta}{\sqrt{2}\cos\theta + \sqrt{2}\sin\theta}\right) = \tan^{-1}\left(\frac{1-\tan\theta}{1+\tan\theta}\right) \tag{66}$$

$$= \tan^{-1}\left(\tan\left(\frac{\pi}{4} - \theta\right)\right) = \frac{\pi}{4} - \theta = \frac{\pi}{4} - \frac{1}{2}\cos^{-1} x \tag{67}$$

Differentiating with respect to x:

$$\frac{dy}{dx} = -\frac{1}{2}\left(-\frac{1}{\sqrt{1-x^2}}\right) = \frac{1}{2\sqrt{1-x^2}} \tag{68}$$

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codes/q15.py
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1.2.16 Question 16 Find the equation of the tangent to the curve $x = \sin 3t$, $y = \cos 2t$ at $t = \frac{\pi}{4}$.
Solution: At $t = \frac{\pi}{4}$:

$$x_0 = \sin\left(\frac{3\pi}{4}\right) = \frac{1}{\sqrt{2}}, \quad y_0 = \cos\left(\frac{2\pi}{4}\right) = \cos\left(\frac{\pi}{2}\right) = 0 \tag{69}$$

$$\frac{dx}{dt} = 3\cos 3t \implies \left.\frac{dx}{dt}\right|_{t=\frac{\pi}{4}} = 3\cos\left(\frac{3\pi}{4}\right) = -\frac{3}{\sqrt{2}} \tag{70}$$

$$\frac{dy}{dt} = -2\sin 2t \implies \left.\frac{dy}{dt}\right|_{t=\frac{\pi}{4}} = -2\sin\left(\frac{\pi}{2}\right) = -2 \tag{71}$$

$$m = \frac{dy/dt}{dx/dt} = \frac{-2}{-3/\sqrt{2}} = \frac{2\sqrt{2}}{3} \tag{72}$$

The equation of the tangent line is:

$$y - 0 = \frac{2\sqrt{2}}{3}\left(x - \frac{1}{\sqrt{2}}\right) \implies 3y = 2\sqrt{2}x - 2 \implies 2\sqrt{2}x - 3y - 2 = 0 \tag{73}$$

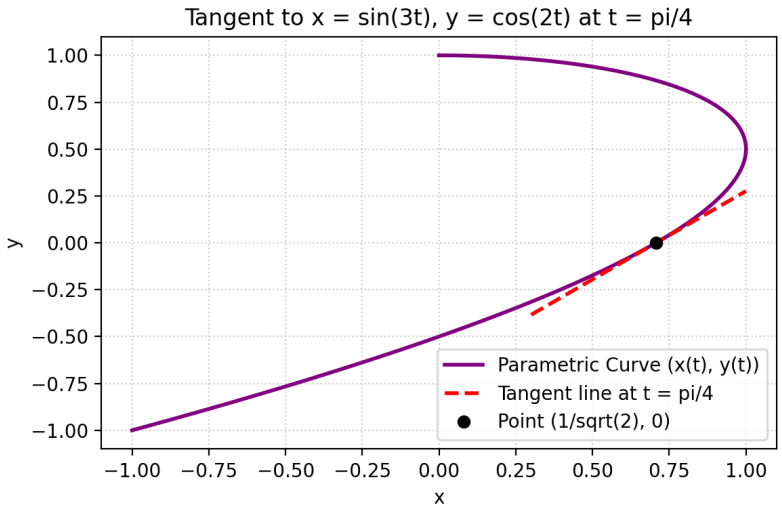


Figure 2: Curve $x = \sin 3t, y = \cos 2t$ and tangent line at $t = \pi/4$.

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codes/q16.py
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1.2.17 Question 17 Evaluate:

$$I = \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x}\, dx \tag{74}$$

Solution: Using formula (20):

$$I = \int_0^{\pi} \frac{(\pi - x)\sin(\pi - x)}{1 + \cos^2(\pi - x)}\, dx = \int_0^{\pi} \frac{(\pi - x)\sin x}{1 + \cos^2 x}\, dx \tag{75}$$

Adding both equations:

$$2I = \pi \int_0^{\pi} \frac{\sin x}{1 + \cos^2 x}\, dx \tag{76}$$

$$I = \frac{\pi}{2} \int_0^{\pi} \frac{\sin x}{1 + \cos^2 x}\, dx \tag{77}$$

Substitute $u = \cos x$, $du = -\sin x dx$. At $x = 0 \implies u = 1$; at $x = \pi \implies u = -1$:

$$I = \frac{\pi}{2} \int_1^{-1} \frac{-du}{1 + u^2} = \frac{\pi}{2} \int_{-1}^{1} \frac{du}{1 + u^2} = \frac{\pi}{2} \left[\tan^{-1} u\right]_{-1}^{1} = \frac{\pi}{2} \left(\frac{\pi}{4} - \left(-\frac{\pi}{4}\right)\right) = \frac{\pi^2}{4} \tag{78}$$

`codes/q17.c`

1.2.18 Question 18 Solve the differential equation: $(x^2 - y^2)dx + 2xydy = 0$ given that $y = 1$ when $x = 1$.
Solution (via Laplace Transform): Rewrite as:

$$\frac{dy}{dx} = \frac{y^2 - x^2}{2xy} \tag{79}$$

Let $u = y^2 \implies \frac{du}{dx} = 2y\frac{dy}{dx}$. Thus:

$$x\frac{du}{dx} + u = x^2 \tag{80}$$

Change variable to the logarithmic domain $t = \ln x$ ($x = e^t$). With $w(t) = u(e^t)$, $\frac{du}{dx} = e^{-t}\frac{dw}{dt}$:

$$\frac{dw}{dt} + w = e^{2t}, \quad w(0) = y(1)^2 = 1 \tag{81}$$

Applying Laplace transform (6) and (8):

$$sW(s) - w(0) + W(s) = \frac{1}{s - 2} \tag{82}$$

$$(s + 1)W(s) = 1 + \frac{1}{s - 2} = \frac{s - 1}{s - 2} \tag{83}$$

$$W(s) = \frac{s - 1}{(s + 1)(s - 2)} = \frac{2/3}{s + 1} + \frac{1/3}{s - 2} \tag{84}$$

Taking the inverse Laplace transform:

$$w(t) = \frac{2}{3}e^{-t} + \frac{1}{3}e^{2t} \tag{85}$$

Transforming back via $t = \ln x$:

$$y^2 = \frac{2}{3x} + \frac{x^2}{3} \implies 3xy^2 = 2 + x^3 \implies x^3 - 3xy^2 + 2 = 0 \tag{86}$$

`codes/q18.py`

1.2.19 Question 19 Solve the differential equation: $\cos^2 x \frac{dy}{dx} + y = \tan x$.

Solution (via Laplace Transform): Dividing through by $\cos^2 x$:

$$\frac{dy}{dx} + \sec^2 x \cdot y = \tan x \sec^2 x \tag{87}$$

Let $u = \tan x \implies \frac{du}{dx} = \sec^2 x$. By the chain rule, $\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx} = \sec^2 x \frac{dy}{du}$:

$$\frac{dy}{du} + y = u \tag{88}$$

Applying Laplace transform (6) with respect to u:

$$sY(s) - y(0) + Y(s) = \frac{1}{s^2} \tag{89}$$

$$(s+1)Y(s) = y(0) + \frac{1}{s^2} \tag{90}$$

$$Y(s) = \frac{y(0)}{s+1} + \frac{1}{s^2(s+1)} \tag{91}$$

Using partial fractions, $\frac{1}{s^2(s+1)} = \frac{1}{s^2} - \frac{1}{s} + \frac{1}{s+1}$. Thus:

$$Y(s) = \frac{1}{s^2} - \frac{1}{s} + \frac{y(0)+1}{s+1} \tag{92}$$

$$y(u) = u - 1 + Ce^{-u} \tag{93}$$

where $C = y(0) + 1$. Substituting $u = \tan x$:

$$y(x) = \tan x - 1 + Ce^{-\tan x} \tag{94}$$

codes/q19.py

1.2.20 Question 20 If $\mathbf{a} = \hat{i} + 4\hat{j} + 2\hat{k}$ and $\mathbf{b} = 3\hat{i} - 2\hat{j} + 7\hat{k}$, find a vector $\mathbf{c}$ such that $\mathbf{a} \times \mathbf{c} = \mathbf{b}$ and $\mathbf{a}^\top \mathbf{c} = 1$.

Solution: Using the vector triple product identity $\mathbf{a} \times (\mathbf{a} \times \mathbf{c}) = (\mathbf{a}^\top \mathbf{c})\mathbf{a} - \|\mathbf{a}\|^2 \mathbf{c}$:

$$\mathbf{a} \times \mathbf{b} = (1)\mathbf{a} - \|\mathbf{a}\|^2 \mathbf{c} \tag{95}$$

$$\|\mathbf{a}\|^2 \mathbf{c} = \mathbf{a} - \mathbf{a} \times \mathbf{b} = \mathbf{a} + \mathbf{b} \times \mathbf{a} \tag{96}$$

From formula (10), $\|\mathbf{a}\|^2 = 1^2 + 4^2 + 2^2 = 21$.

$$\mathbf{b} \times \mathbf{a} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -2 & 7 \\ 1 & 4 & 2 \end{vmatrix} = (-4 - 28)\hat{i} - (6 - 7)\hat{j} + (12 + 2)\hat{k} = -32\hat{i} + \hat{j} + 14\hat{k} \tag{97}$$

Therefore:

$$\mathbf{c} = \frac{1}{21}\left[(\hat{i} + 4\hat{j} + 2\hat{k}) + (-32\hat{i} + \hat{j} + 14\hat{k})\right] = \frac{-31\hat{i} + 5\hat{j} + 16\hat{k}}{21} \tag{98}$$

codes/q20.c

1.2.21 Question 21 Find the shortest distance between the lines:

$$\frac{x-3}{1}=\frac{y-5}{-2}=\frac{z-7}{1} \quad \text{and} \quad \frac{x+1}{7}=\frac{y+1}{-6}=\frac{z+1}{1} \tag{99}$$

Solution: Points and direction vectors are:

$$\mathbf{a}_1=\begin{pmatrix}3\\5\\7\end{pmatrix}, \quad \mathbf{b}_1=\begin{pmatrix}1\\-2\\1\end{pmatrix}; \qquad \mathbf{a}_2=\begin{pmatrix}-1\\-1\\-1\end{pmatrix}, \quad \mathbf{b}_2=\begin{pmatrix}7\\-6\\1\end{pmatrix} \tag{100}$$

$$\mathbf{a}_2-\mathbf{a}_1=\begin{pmatrix}-4\\-6\\-8\end{pmatrix} \tag{101}$$

From formula (13):

$$\mathbf{n}=\mathbf{b}_1\times\mathbf{b}_2=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 1 & -2 & 1\\ 7 & -6 & 1\end{vmatrix}=4\hat{i}+6\hat{j}+8\hat{k} \tag{102}$$

By the norm procedure (10):

$$\|\mathbf{n}\|=\sqrt{4^2+6^2+8^2}=\sqrt{16+36+64}=\sqrt{116}=2\sqrt{29} \tag{103}$$

From formula (14):

$$d=\frac{|(-4)(4)+(-6)(6)+(-8)(8)|}{2\sqrt{29}}=\frac{|-16-36-64|}{2\sqrt{29}}=\frac{116}{2\sqrt{29}}=\frac{58}{\sqrt{29}}=2\sqrt{29} \tag{104}$$

codes/q21.c

1.2.22 Question 22 A pair of dice is thrown 4 times. If getting a doublet is considered a success, find the probability distribution of the number of successes.

Solution: When a pair of dice is rolled, the sample space has 36 outcomes. Doublets are $\{(1,1),(2,2),(3,3),(4,4),(5,5),(6$ (6 outcomes).

$$p=\frac{6}{36}=\frac{1}{6}, \quad q=1-p=\frac{5}{6}, \quad n=4 \tag{105}$$

From Binomial distribution formula (4):

$$P(X=0)=\binom{4}{0}\left(\frac{1}{6}\right)^0\left(\frac{5}{6}\right)^4=\frac{625}{1296} \tag{106}$$

$$P(X=1)=\binom{4}{1}\left(\frac{1}{6}\right)^1\left(\frac{5}{6}\right)^3=\frac{500}{1296}=\frac{125}{324} \tag{107}$$

$$P(X=2)=\binom{4}{2}\left(\frac{1}{6}\right)^2\left(\frac{5}{6}\right)^2=\frac{150}{1296}=\frac{25}{216} \tag{108}$$

$$P(X=3)=\binom{4}{3}\left(\frac{1}{6}\right)^3\left(\frac{5}{6}\right)^1=\frac{20}{1296}=\frac{5}{324} \tag{109}$$

$$P(X=4)=\binom{4}{4}\left(\frac{1}{6}\right)^4\left(\frac{5}{6}\right)^0=\frac{1}{1296} \tag{110}$$

codes/q22.c

Section C (6 Marks Each)

1.2.23 Question 23 Using properties of determinants, prove that:

$$\begin{vmatrix} \alpha & \beta & \gamma \\ \alpha^2 & \beta^2 & \gamma^2 \\ \beta+\gamma & \gamma+\alpha & \alpha+\beta \end{vmatrix} = (\beta-\gamma)(\gamma-\alpha)(\alpha-\beta)(\alpha+\beta+\gamma) \tag{111}$$

Solution: Applying $R_3 \to R_3 + R_1$:

$$\Delta = \begin{vmatrix} \alpha & \beta & \gamma \\ \alpha^2 & \beta^2 & \gamma^2 \\ \alpha+\beta+\gamma & \alpha+\beta+\gamma & \alpha+\beta+\gamma \end{vmatrix} = (\alpha+\beta+\gamma)\begin{vmatrix} \alpha & \beta & \gamma \\ \alpha^2 & \beta^2 & \gamma^2 \\ 1 & 1 & 1 \end{vmatrix} \tag{112}$$

Applying column operations $C_1 \to C_1 - C_2$ and $C_2 \to C_2 - C_3$:

$$\Delta = (\alpha+\beta+\gamma)\begin{vmatrix} \alpha-\beta & \beta-\gamma & \gamma \\ \alpha^2-\beta^2 & \beta^2-\gamma^2 & \gamma^2 \\ 0 & 0 & 1 \end{vmatrix} \tag{113}$$

$$= (\alpha+\beta+\gamma)(\alpha-\beta)(\beta-\gamma)\begin{vmatrix} 1 & 1 & \gamma \\ \alpha+\beta & \beta+\gamma & \gamma^2 \\ 0 & 0 & 1 \end{vmatrix} \tag{114}$$

$$= (\alpha+\beta+\gamma)(\alpha-\beta)(\beta-\gamma)[(\beta+\gamma)-(\alpha+\beta)] \tag{115}$$

$$= (\alpha+\beta+\gamma)(\alpha-\beta)(\beta-\gamma)(\gamma-\alpha) \tag{116}$$

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codes/q23.c
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1.2.24 Question 24 Show that the rectangle of maximum area that can be inscribed in a circle of radius R is a square.

Solution: Let length be $2x$ and breadth be $2y$. The circle equation is $x^2+y^2=R^2 \implies y=\sqrt{R^2-x^2}$.

$$\text{Area } A = (2x)(2y) = 4x\sqrt{R^2-x^2} \tag{117}$$

Maximizing $S = A^2 = 16(x^2R^2 - x^4)$:

$$\frac{dS}{dx} = 16(2xR^2 - 4x^3) = 0 \implies 2x^2 = R^2 \implies x = \frac{R}{\sqrt{2}} \tag{118}$$

Then $y = \sqrt{R^2 - \frac{R^2}{2}} = \frac{R}{\sqrt{2}}$. Furthermore, $\frac{d^2S}{dx^2} = 16(2R^2 - 12x^2) = -64R^2 < 0$. Since $x = y$, length = breadth $= \sqrt{2}R$. Hence, the rectangle of maximum area is a square.

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codes/q24.c
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1.2.25 Question 25 Using integration, find the area of the region bounded by the parabola $y^2 = 4x$ and the circle $4x^2 + 4y^2 = 9$.

Solution: Substituting $y^2 = 4x$ into $4x^2 + 4y^2 = 9$:

$$4x^2 + 16x - 9 = 0 \implies (2x-1)(2x+9) = 0 \implies x = \frac{1}{2} \quad (x \geq 0) \tag{119}$$

Due to symmetry across the x-axis, the total area is:

$$A = 2\left[\int_0^{1/2} 2\sqrt{x}\,dx + \int_{1/2}^{3/2} \sqrt{\frac{9}{4} - x^2}\,dx\right] \tag{120}$$

$$= 4\left[\frac{2}{3}x^{3/2}\right]_0^{1/2} + 2\left[\frac{x}{2}\sqrt{\frac{9}{4} - x^2} + \frac{9}{8}\sin^{-1}\left(\frac{2x}{3}\right)\right]_{1/2}^{3/2} \tag{121}$$

$$= \frac{\sqrt{2}}{6} + \frac{9\pi}{8} - \frac{9}{4}\sin^{-1}\left(\frac{1}{3}\right) \tag{122}$$

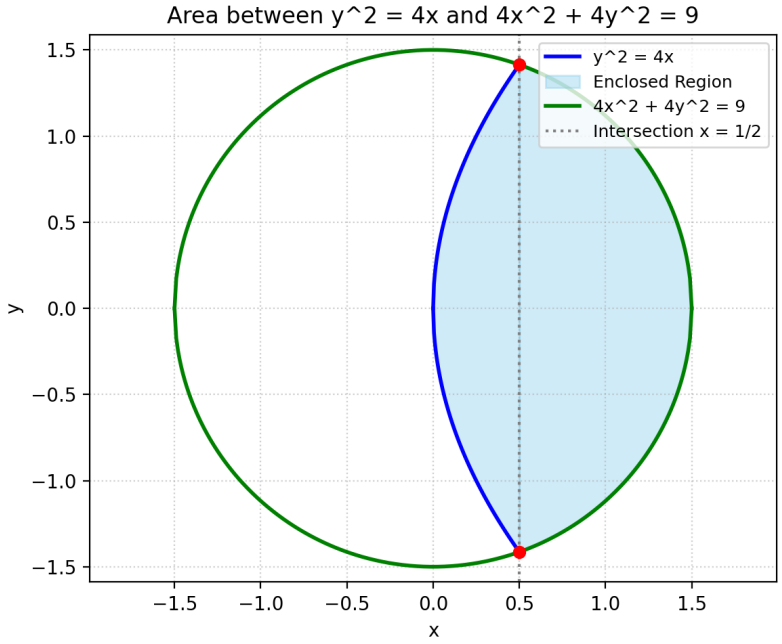


Figure 3: Region bounded by parabola $y^2 = 4x$ and circle $4x^2 + 4y^2 = 9$.

codes/q25.py

1.2.26 Question 26 Evaluate:

$$I = \int_{-a}^{a} \sqrt{\frac{a-x}{a+x}}\,dx \tag{123}$$

Solution: Rationalizing the numerator:

$$I = \int_{-a}^{a} \frac{a-x}{\sqrt{a^2-x^2}}\,dx = a\int_{-a}^{a} \frac{dx}{\sqrt{a^2-x^2}} - \int_{-a}^{a} \frac{x}{\sqrt{a^2-x^2}}\,dx \tag{124}$$

The second integral has an odd integrand over $[-a, a]$, so it evaluates to 0. The first integrand is even:

$$I = 2a\int_0^{a} \frac{dx}{\sqrt{a^2-x^2}} = 2a\left[\sin^{-1}\left(\frac{x}{a}\right)\right]_0^{a} = 2a\left(\frac{\pi}{2} - 0\right) = \pi a \tag{125}$$

codes/q26.c

1.2.27 Question 27 Find the equation of the plane passing through $(-1,-1,2)$ and perpendicular to the planes $2x+3y-3z=2$ and $5x-4y+z=6$.

Solution: Normal vectors to the given planes are $\mathbf{n}_1 = \begin{pmatrix} 2 \\ 3 \\ -3 \end{pmatrix}$ and $\mathbf{n}_2 = \begin{pmatrix} 5 \\ -4 \\ 1 \end{pmatrix}$. The normal $\mathbf{n}$ to the required plane is:

$$\mathbf{n} = \mathbf{n}_1 \times \mathbf{n}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & -3 \\ 5 & -4 & 1 \end{vmatrix} = -9\hat{i} - 17\hat{j} - 23\hat{k} \tag{126}$$

Using point-normal form with point $\mathbf{x}_0 = (-1,-1,2)^\top$:

$$-9(x+1) - 17(y+1) - 23(z-2) = 0 \tag{127}$$

$$-9x - 17y - 23z - 9 - 17 + 46 = 0 \tag{128}$$

$$9x + 17y + 23z = 20 \tag{129}$$

codes/q27.c

1.2.28 Question 28 A factory owner purchases machines A and B. Area occupied: A needs $1000\,\mathrm{m}^2$, B needs $1200\,\mathrm{m}^2$ (Max $9000\,\mathrm{m}^2$). Labour required: A needs 12 men, B needs 8 men (Max 72 men). Daily output: A produces 60, B produces 40. How many machines of each type maximize daily output?

Solution: Let x be the number of machine A and y be the number of machine B.

$$\text{Maximize } Z = 60x + 40y \tag{130}$$

$$\text{Subject to } 1000x + 1200y \leq 9000 \implies 5x + 6y \leq 45 \tag{131}$$

$$12x + 8y \leq 72 \implies 3x + 2y \leq 18 \tag{132}$$

$$x, y \geq 0, \quad x, y \in \mathbb{Z}_{\geq 0} \tag{133}$$

The corner points of the feasible region are:

- $(0,0) \implies Z = 0$
- $(6,0) \implies Z = 60(6) + 40(0) = 360$
- $(0,7) \implies Z = 60(0) + 40(7) = 280$
- Feasible integer points near the boundary: $(2,5) \implies 3(2) + 2(5) = 16 \leq 18$, $5(2) + 6(5) = 40 \leq 45 \implies Z = 120 + 200 = 320$.

Maximum daily output is 360 units, achieved by purchasing 6 machines of type A and 0 machines of type B.

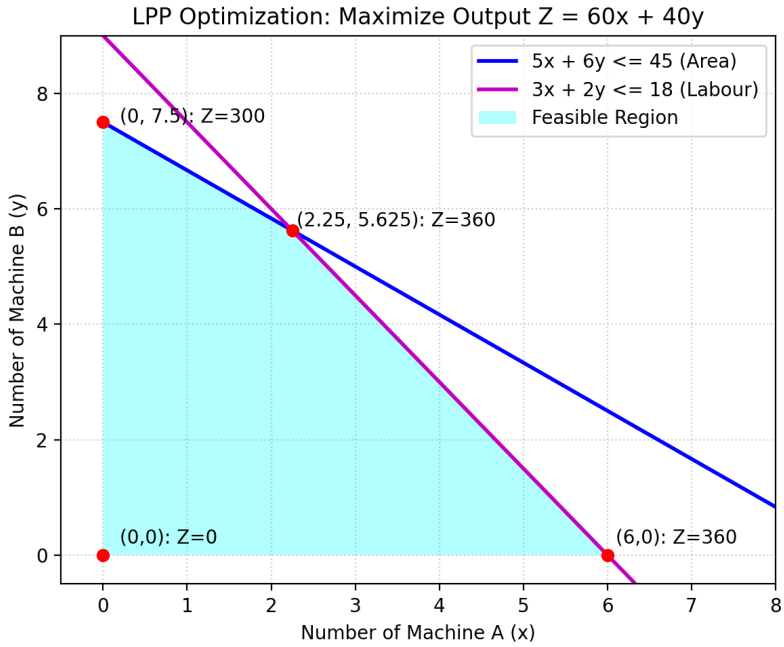


Figure 4: Feasible region and optimal corner point for machine allocation.

codes/q28.py

1.2.29 Question 29 An insurance company insured 2000 scooter drivers, 4000 car drivers, and 6000 truck drivers. Probability of accident: scooter = 0.01, car = 0.03, truck = 0.15. An insured person meets with an accident. What is the probability that he is a scooter driver?

Solution: Total insured drivers $= 2000 + 4000 + 6000 = 12000$. Let E_1, E_2, E_3 be the events of selecting scooter, car, and truck drivers respectively, and A be the event of an accident:

$$\Pr(E_1) = \frac{2000}{12000} = \frac{1}{6}, \quad \Pr(A|E_1) = 0.01 = \frac{1}{100} \tag{134}$$

$$\Pr(E_2) = \frac{4000}{12000} = \frac{1}{3}, \quad \Pr(A|E_2) = 0.03 = \frac{3}{100} \tag{135}$$

$$\Pr(E_3) = \frac{6000}{12000} = \frac{1}{2}, \quad \Pr(A|E_3) = 0.15 = \frac{15}{100} \tag{136}$$

Applying Bayes' Theorem (3):

$$\Pr(E_1|A) = \frac{\Pr(E_1)\Pr(A|E_1)}{\Pr(E_1)\Pr(A|E_1) + \Pr(E_2)\Pr(A|E_2) + \Pr(E_3)\Pr(A|E_3)} \tag{137}$$

$$= \frac{\frac{1}{6} \times \frac{1}{100}}{\frac{1}{6} \times \frac{1}{100} + \frac{1}{3} \times \frac{3}{100} + \frac{1}{2} \times \frac{15}{100}} \tag{138}$$

$$= \frac{\frac{1}{6}}{\frac{1}{6} + 1 + \frac{15}{2}} = \frac{\frac{1}{6}}{\frac{1+6+45}{6}} = \frac{1}{52} \tag{139}$$

codes/q29.c